补充与总结
Supplementary and Summary

现代C++基础
Modern C++ Basics

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- **File system**
- **Time utilities (Chrono)**
- **Math utilities**
- **Summary and future prospect**

Supplementary and Summary

File system

Overview

- Files represent organized data in non-volatile storage to let programs share data across different runs.
  - Files are named collection of data.
- Directories are used to construct file hierarchy.
  - Directories are named collection of files and directories.

- File system is an abstraction layer provided by OS to enable users to use *path* to access files and directories.
  - It records metadata of files and directories (size, modification time, owner, etc.) to make them organized and hierarchical.
- C++17 includes related utilities in `<filesystem>`.

Supplementary

- File system
  - Path operations
    - Overview
    - std::filesystem::path
  - File system operations

Path Overview

- Essentially, path is a string that represents location of a file.
- There are two different kinds of paths:
  1. Absolute path: always refer to the same location.
  2. Relative path: the location relative to *current working directory* (CWD) for the current process.
     - By changing CWD, the process can get different locations.
- A path consists of these components:
  1. Root name (optional): like drive name in Windows (C:, D:); or UNC (`//machine`), etc.
  2. Root directory (optional): a directory separator (`\` on Windows, `/` on Linux, `:` on classic MacOS).
  3. Relative path: a sequence of filenames separated by directory separator.

Path Overview

- Besides no separators, filenames have many other platform-dependent characteristics or restrictions.

(1.1) — The permitted characters. e.g. on Windows: 文件名不能包含下列任何字符: \ / : * ? " < > |

[*Example 1*: Some operating systems prohibit the ASCII control characters (0x00 – 0x1F) in filenames. — *end example*]

[*Note 1*: Wider portability can be achieved by limiting *filename* characters to the POSIX Portable Filename Character Set:
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
0 1 2 3 4 5 6 7 8 9 . _ - — *end note*]

(1.2) — The maximum permitted length. e.g. well-known 260 in some Windows functions.

(1.3) — Filenames that are not permitted.

(1.4) — Filenames that have special meaning. e.g. CON on Windows.

(1.5) — Case awareness and sensitivity during path resolution. e.g. Linux is case-sensitive while Windows not.

(1.6) — Special rules that may apply to file types other than regular files, such as directories.

Particularly, . and .. means current directory and parent directory respectively.

Path Overview

- To make program cross-platform, C++ regulates a “generic format” that uses POSIX convention.
  - That is, these three components are just concatenated to form a path.
    - And / is considered as universal separator.
- Besides, C++ allows “native format” that depends on file system.
  - E.g. OpenVMS, a legacy system previously used in bank (well, if it really supports C++17 utilities).

What Is a Fully Qualified Name?

A *fully qualified name* indicates how a file fits into a structure (a system of directories and subdirectories) that contains all the files stored under the OpenVMS system. The following type of file specification is a fully qualified name:

node::*device*:[*directory*]*filename*.*file-type*;*version* e.g. DKA0:[JDOE.DATA]test.txt

Path Overview

- Note 1: “generic” only means it’s a valid format in all systems; but its location is not always the same.
  - What is `D:\sub\path`?
    - Windows: an absolute path at D drive, with two components `sub` and `path`.
    - Linux: a relative path with name “`D:\sub\path`”, i.e. the whole string is a single component.
  - What is `/home/user`?
    - Linux: an absolute path with two components `home` and `user`.
    - Windows: a relative path at drive of CWD, with two components `home` and `user`.
- Note 2: it’s not very safe to rely on relative path since it’s just like a global variable, which can be changed by other threads and external library.
  - i.e. the location of a relative path can be modified arbitrarily.

Path Overview

- Note 3: “relative” or “absolute” only means whether it’s interfered by CWD; there can be many paths that refer to the same location.
  - E.g. `/home/user`, `/home/user/.`, `/home/user/dir/..`, ….
  - To unify all representations, we can do *path normalization*.
  - There are two kinds of normalization:
    - Lexical: a string-level substitution, which doesn’t change whether the path is relative or absolute.
      - So `/home/user`, `/home/user/.`, `/home/user/dir/..` are all normalized to `/home/user`, and paths `./user`, `./user/.`, `./user/dir/..` are all normalized to `user`.
    - Filesystem-dependent: paths are normalized to a unique absolute path.
      - Assuming CWD is `/home`, `./user`, `./user/.`, `./user/dir/..` are all normalized to `/home/user`.
      - C++ call such normalization as “canonical”.

Path Overview

- Specifically, a normalized path requires:

A path might be or can become normalized. In a normalized path:

- File names are separated only by a single preferred directory separator.
- The file name `".."` is not used unless the whole path is nothing but `".."` (representing the current directory).
- The file name does not contain `"..."` file names (we do not go down and then up again) unless they are at the beginning of a relative path.
- The path only ends with a directory separator if the trailing file name is a directory with a name other than `".."` or `"..."`.

Path	POSIX normalized	Windows normalized
`foo/.///bar/../`	`foo/`	`foo\`
`//host/../foo.txt`	`//host/foo.txt` POSIX may or may not respect //.	`\\host\foo.txt` (Due to UNC path)
`./f/../.f/`	`.f/`	`.f\`
`C:bar/../`	`.`	`C:`
`C:/bar/..`	`C:/`	`C:\`
`C:\bar\..`	`C:\bar\..`	`C:\`
`/./../data.txt`	`/data.txt`	`\data.txt`
`././`	`.`	`.`

Credit: C++17 the Complete Guide, *Nicolai M. Josuttis*.

Supplementary

- File system
  - Path operations
    - Overview
    - std::filesystem::path
  - File system operations

For brevity, we use `namespace stdfs = std::filesystem`.

Path

- C++ uses `stdfs::path` to represent a path.
  - It is essentially a string of some *native encoding*, e.g. UTF-8 on Linux, UTF-16 on Windows*.
    - The underlying character can be checked by `stdfs::path::value_type`, normally `char` on Linux and `wchar_t` on Windows.
    - And `stdfs::path::string_type = std::basic_string<value_type>`.
  - And the string stores path in *native format*.
  - You can just access the underlying string in `const` way:

Accesses the native path name as a character string.

`const value_type* c_str() const noexcept;`	(1)	1) Equivalent to `native().c_str()`.
`const string_type& native() const noexcept;`	(2)	2) Returns the native-format representation of the pathname by reference.
`operator string_type() const;`	(3)	3) Returns the native-format representation of the pathname by value.

*: Strictly speaking, usually a filesystem doesn't really respect encoding; it just treats the path as some byte sequence (even if it's not a valid UTF-8 / 16). So "native encoding" essentially means "a string that you can pass into filesystem syscall directly", e.g. 2-byte-per-unit sequence on Windows.

Path

Exceptions

2,4-8) May throw implementation-defined exceptions.

- However, you can construct the path by any encoding and format.

```
template< class Source >
path( const Source& source, format fmt = auto_format );
```

```
template< class InputIt >
path( InputIt first, InputIt last, format fmt = auto_format );
```

- For `format`, it's essentially a scoped enumeration in `stdfs::path` with three enumerators:

Name	Explanation
`native_format`	native pathname format
`generic_format`	generic pathname format
`auto_format`	implementation-defined format, auto-detected where possible

- By default it uses `auto_format`, i.e. determine if the input format is native or generic automatically and convert if necessary.

Path

```
template< class Source >
path( const Source& source, format fmt = auto_format );
```

```
template< class InputIt >
path( InputIt first, InputIt last, format fmt = auto_format );
```

- The template allows you to use any character type, and ctor will convert to its native encoding.

(2.1) — `char`: The encoding is the native ordinary encoding. The method of conversion, if any, is operating system dependent.

(2.2) — `wchar_t`: The encoding is the native wide encoding. The method of conversion is unspecified.

(2.3) — `char8_t`: The encoding is UTF-8. The method of conversion is unspecified.

(2.4) — `char16_t`: The encoding is UTF-16. The method of conversion is unspecified.

(2.5) — `char32_t`: The encoding is UTF-32. The method of conversion is unspecified.

  - For `wchar_t`, Windows won't do any conversion but Linux needs to do so;
  - For `char`, Linux won't do any conversion but Windows needs to do so.
- Actually, Windows recognizes `char` for file API in ANSI (or Active) Code Page (ACP).
  - This will lead to complex behavior when interacting with compiler option…

Windows ACP

- Assuming that we have a file path "`D:\试验.txt`" on Windows.
  - And we use a default Chinese PC, i.e. ACP is GBK (id: 936).
- Given `main.cpp` as:

```
std::filesystem::path p{ R"(D:\试验.txt)" };
std::cout << std::filesystem::exists(p) << "\n";
```

Check whether some path exists, equiv. to `std::ifstream{p}.is_open()` if `p` is a file instead of directory.

- Case 1: msvc doesn't add any option, and encoding of `main.cpp` is GBK.
  1. `D:\试验.txt` is in GBK, and msvc reads it as GBK correctly.
  2. The execution charset is GBK, so `D:\试验.txt` is still GBK in binary exe.
  3. Current ACP is GBK, so `stdfs::path` converts it from GBK to native encoding (UTF-16) and stores it;
  4. The path is correct so file system says it exists.

Windows ACP

```
std::filesystem::path p{ R"(D:\试验.txt)" };
std::cout << std::filesystem::exists(p) << "\n";
```

- Case 2: msvc adds `/utf-8`, and encoding of `main.cpp` is UTF-8.
  1. `D:\试验.txt` is in UTF-8, and msvc reads it as UTF-8 correctly.
  2. The execution charset is UTF-8, so `D:\试验.txt` **is UTF-8** in binary exe.
  3. Current ACP is GBK, so `stdfs::path` converts it **from GBK** to native encoding (UTF-16) and stores it;
     - However, UTF-8 string is not really a GBK string, and the corresponding binary leads to GBK as "`D:\璇瞭獙.txt`".
  4. The path is not correct and file system says it doesn't exist.
- Case 3: msvc adds `/source-charset:utf-8`, and encoding of `main.cpp` is UTF-8.
  1. `D:\试验.txt` is in UTF-8, and msvc reads it as UTF-8 correctly.
  2. As default execution charset is ACP, `D:\试验.txt` **is GBK** in binary exe.
  3. So the path is still correct and file system says it exists.

Windows ACP

```
std::filesystem::path p{ R"(D:\试验.txt)" };
std::cout << std::filesystem::exists(p) << "\n";
```

- Case 4: msvc adds `/utf-8`, and encoding of `main.cpp` is GBK.
  1. `D:\试验.txt` in GBK is not valid UTF-8, so msvc warns C4828 (illegal character in UTF-8) and silently passes the original bytes as is.
     - So **accidentally**, it's still GBK in binary exe.
  2. Thus accidentally, the path is correct and file system says it exists.
- Case 5: assuming ACP is UTF-8 (65001), msvc doesn't add any option (equiv. to add `/utf-8`), and encoding of `main.cpp` is GBK.
  1. Same as case 4, i.e. the string is of GBK in binary exe.
  2. However, GBK is not valid UTF-8, so ctor of path throws an exception (`std::system_error` in MS-STL).

Path construction

- So to make a valid path, we need first:
  - Make sure the compiler knows how to read the file (string literals), i.e. the file encoding should be correctly specified in source charset.
- And then two ways:
  1. Make execution charset (for string literals) and string encoding (for other strings stored in e.g. `std::string`) same as ACP.
  2. Use `char8_t[] / char16_t[] / char32_t[]` instead.
     A. Any ACP is OK, since `char8_t` as a unique type will always be decoded as UTF-8 in ctor.
     B. Any execution charset is OK, since `char8_t` is always UTF-8 in binary exe.

```
std::filesystem::path p{ u8R"(D:\试验.txt)" };
```

- And if you know your `std::string / …` is essentially UTF-8 while ACP is not, you can `reinterpret_cast` it to avoid conversion.

```
// Assuming we use UTF-8 as execution charset.
std::string s{ R"(D:\试验.txt)" };
auto ptr = reinterpret_cast<const char8_t *>(s.c_str());
std::u8string_view view{ ptr, ptr + s.size() / sizeof(char8_t) };
std::filesystem::path p{ view };
```

Path construction

- On the other hand, on Linux + gcc, no matter what execution charset you use, `char[]` won't do any conversion.
  - As `char` is its native encoding, so libstdc++ assumes correct bytes.
  - Instead, `-fwide-exec-charset` will specify encoding of `wchar_t` and be converted to UTF-8 automatically.
- To specify encoding and conversion explicitly, you can use locale:

```
template< class Source >
path( const Source& source, const std::locale& loc, format fmt = auto_format );
```

```
template< class InputIt >
path( InputIt first, InputIt last, const std::locale& loc, format fmt = auto_format );
```

  - As the conversion is explicitly specified in locale, the template only accepts a char sequence.

Path const

```
path& operator=( const path& p );
path& operator=( path&& p ) noexcept;
path& operator=( string_type&& source );
template< class Source >
path& operator=( const Source& source );
```

```
path& assign( string_type&& source );
template< class Source >
path& assign( const Source& source );
template< class InputIt >
path& assign( InputIt first, InputIt last );
```

- When native encoding is `wchar_t`:
  - Just use `codecvt<wchar_t, char, std::mbstate_t>` in locale to convert `char[]` to native encoding `wchar_t[]`.
    - E.g. For windows, GBK -> UTF-16.
- When native encoding is `char`:
  - First use `codecvt<wchar_t, char, std::mbstate_t>` in locale to convert `char[]` to `wchar_t[]`;
    - E.g. For Linux, GBK -> UTF-32
  - Then convert `wchar_t` back to native encoding `char` (e.g. UTF-8).
    - E.g. UTF-32 -> UTF-8, which is equiv. to construct from `wchar_t[]` directly.

- And finally some omitted overloads & `operator=`, just list here.

```
path() noexcept;
```

Default is just an empty string.

```
path( const path& p );
path( path&& p ) noexcept;
path( string_type&& source, format fmt = auto_format );
```

Path

- Besides construction, you can also get string of different formats and encodings.
  - Native format + Native encoding: just `.native()`, as we mentioned.
  - Native format + Converted encoding:

```
std::string string() const;
std::wstring wstring() const;
std::u16string u16string() const;
std::u32string u32string() const;

std::u8string u8string() const;
```

  - Generic format + Converted encoding:

```
std::string generic_string() const;
std::wstring generic_wstring() const;
std::u16string generic_u16string() const;
std::u32string generic_u32string() const;
```
(2) (since C++17)

```
std::u8string generic_u8string() const;
```
(3) (since C++20)

    - Particularly, these functions are required to use `/` as directory separator.

Path

- And there are also template versions, together will allocator:

```
template< class CharT, class Traits = std::char_traits<CharT>,
          class Alloc = std::allocator<CharT> >
std::basic_string<CharT,Traits,Alloc>
    string( const Alloc& a = Alloc() ) const;
```
(1)

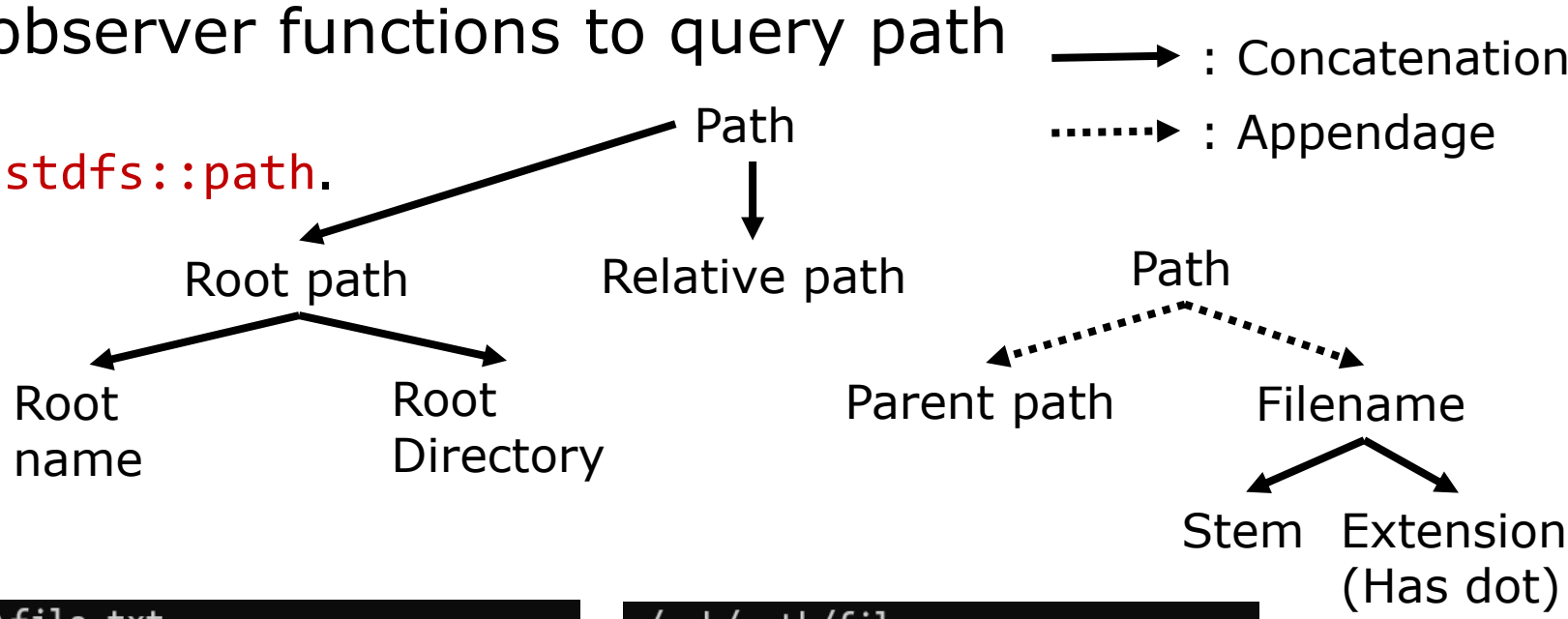
```
template< class CharT, class Traits = std::char_traits<CharT>,
          class Alloc = std::allocator<CharT> >
std::basic_string<CharT,Traits,Alloc>
    generic_string( const Alloc& a = Alloc() ) const;
```
(1)

  - So to get Generic format + Native encoding, just use `.generic_string<stdfs::path::value_type>()`.
- Finally, you can also check the preferred separator by `static constexpr stdfs::path::preferred_separator`.
  - `\` on Windows, `/` on Linux.

Path decomposition

Note: filename with pure extension (e.g. "D:\\.gitignore") is not considered as extension (i.e. stem = filename = ".gitignore", extension = "")

- There are also some observer functions to query path components:
  - which just return new `stdfs::path`.



`root_name`
`root_directory`
`root_path`
`relative_path`
`parent_path`
`filename`
`stem`
`extension`

```
D:\sub\path\file.txt
Original path = "D:\\sub\\path\\file.txt"
Root name = "D:"
Root directory = "\\"
Root path = "D:\\"
Relative path = "sub\\path\\file.txt"
Parent path = "D:\\sub\\path"
Filename = "file.txt"
Stem = "file"
Extension = ".txt"
```

```
./sub/path/file
Original path = "./sub/path/file"
Root name = ""
Root directory = ""
Root path = ""
Relative path = "./sub/path/file"
Parent path = "./sub/path"
Filename = "file"
Stem = "file"
Extension = ""
```

Path decomposition

- Particularly, such decomposition is **lexical**, which doesn't even really interact with file system.
  - So "parent path" doesn't really return path of parent directory, but just remove the last component.
    - And when a path ends with directory separator, the last component is just empty, so parent just removes the separator.
  - For example:

```
D:\sub\path\file\
Original path = "D:\\sub\\path\\file\\"
Root name = "D:"
Root directory = "\\"
Root path = "D:\\"
Relative path = "sub\\path\\file\\"
Parent path = "D:\\sub\\path\\file"
Filename = ""
Stem = ""
Extension = ""
```

```
./sub/path/..
Original path = "./sub/path/.."
Root name = ""
Root directory = ""
Root path = ""
Relative path = "./sub/path/.."
Parent path = "./sub/path"
Filename = ".."
Stem = ".."
Extension = ""
```

Note: parent of root directory is still root directory (i.e. `"/"` -> `"/"`);
but parent of file name is empty (i.e. `"data.txt"` -> `""`).

Path normalization

- To find real parent, you need to do path normalization first.
  - And we say that there are two ways:
  - Lexical: by `.lexically_normal()`; the normalization process is:

Assuming we have a path as "D:/..\sub\.\/path\..\file.txt".

Normalization of a generic format pathname means:

1. If the path is empty, stop.
2. Replace each slash character in the *root-name* with a *preferred-separator*. — D:/..\sub\.\/path\..\file.txt
3. Replace each *directory-separator* with a *preferred-separator*. — D:\..\sub\.\/path\..\file.txt

   [*Note 4*: The generic pathname grammar defines *directory-separator* as one or more slashes and *preferred-separators*. — *end note*]

4. Remove each dot filename and any immediately following *directory-separator*. — D:\..\sub\.\path\..\file.txt
5. As long as any appear, remove a non-dot-dot filename immediately followed by a *directory-separator* and a dot-dot filename, along with any immediately following *directory-separator*. — D:\..\sub\path\..\file.txt
6. If there is a *root-directory*, remove all dot-dot filenames and any *directory-separators* immediately following them. — D:\..\sub\file.txt

   [*Note 5*: These dot-dot filenames attempt to refer to nonexistent parent directories. — *end note*]

   D:\sub\file.txt
7. If the last filename is dot-dot, remove any trailing *directory-separator*.
8. If the path is empty, add a dot.

7. is applied to e.g. ..\..\.

1. & 8. An empty path is still empty after normalization, but a non-empty but essentially empty is normalized to ..

Path normalization

- Filesystem-dependent: `stdfs::canonical / weakly_canonical`; paths are normalized to a unique absolute path.
  - Path is first converted to an absolute path by `stdfs::absolute(p)`;
  - Then perform lexical normalization.

  - `canonical` will check whether the path really exists, while `weakly_canonical` just normalizes it.
  - We'll go into details about two forms of global APIs in `stdfs` later.

```
path canonical( const std::filesystem::path& p );                    (1)
path canonical( const std::filesystem::path& p,
                std::error_code& ec );                               (2)
path weakly_canonical( const std::filesystem::path& p );             (3)
path weakly_canonical( const std::filesystem::path& p,
                       std::error_code& ec );                        (4)
```

```
std::filesystem::path p{ s };
std::cout << "Original path = " << p << "\n";
std::cout << "-------------- Lexical normal --------------\n";
OutputProperties(p.lexically_normal());
std::cout << "-------------- Weakly canonical --------------\n";
OutputProperties(std::filesystem::weakly_canonical(p));
```

Example

```
Original path = "./sub/path/.."
-------------- Lexical normal --------------
Path = "sub/"
Root name = ""
Root directory = ""
Root path = ""
Relative path = "sub/"
Parent path = "sub"
Filename = ""
Stem = ""
Extension = ""
-------------- Weakly canonical --------------
Path = "/app/sub/"
Root name = ""
Root directory = "/"
Root path = "/"
Relative path = "app/sub/"
Parent path = "/app/sub"
Filename = ""
Stem = ""
Extension = ""
```

The last / is not stripped, even after normalization. .. only removes **trailing** /.

```
Original path = "./sub/path/"
-------------- Lexical normal --------------
Path = "sub/path/"
Root name = ""
Root directory = ""
Root path = ""
Relative path = "sub/path/"
Parent path = "sub/path"
Filename = ""
Stem = ""
Extension = ""
-------------- Weakly canonical --------------
Path = "/app/sub/path/"
Root name = ""
Root directory = "/"
Root path = "/"
Relative path = "app/sub/path/"
Parent path = "/app/sub/path"
Filename = ""
Stem = ""
Extension = ""
```

Path normalization

- Notice that "physical" parent of lexical normalization may still be wrong when normalized result still contains `..`.
  - Only `(weakly_)canonical` ensures a correct physical parent.

```
Original path = "../.."
-------------- Lexical normal --------------
Path = "../.."
Root name = ""
Root directory = ""
Root path = ""
Relative path = "../.."
Parent path = ".."
Filename = ".."
Stem = ".."
Extension = ""
```

- Finally, you can get or set CWD by `current_path()`:

```
std::cout << std::filesystem::current_path() << "\n";
std::filesystem::current_path("C:\\Temp");
std::cout << std::filesystem::current_path() << "\n";
```

```
"D:\\Work\\C++\\Tests\\Project1"
"C:\\Temp"
```

  - So `absolute()` is essentially `current_path() / path` when `path` is relative.

DOS directory

```
C:\Users\Public>cd D:\Work

C:\Users\Public>D:

D:\Work>C:

C:\Users\Public>
```

cd: change current directory

"D:" : switch to current directory of Drive D.

"C:" : switch to current directory of Drive C.

- It's also worth nothing that DOS maintains separate "current directory" for every drive.
  - So `C:` and `D:` are actually relative paths, while `C:\` and `D:\` are absolute paths.
- In Windows, current directory is unified as a single path, as known by CWD.
  - However, CMD pretends they are still there by storing them with "strange environment variables".
  - And Windows inherits the DOS behavior, regarding `C:` and `D:` as relative paths. But there exists only a single real CWD.

```
std::cout << std::filesystem::current_path() << "\n";
std::cout << std::filesystem::absolute("C:") << "\n";
std::cout << std::filesystem::absolute("D:") << "\n";
std::filesystem::current_path("C:\\Temp");
std::cout << std::filesystem::absolute("C:") << "\n";
std::cout << std::filesystem::absolute("D:") << "\n";
```

```
"D:\\Work\\C++\\Tests\\Project1"
"C:\\"
"D:\\Work\\C++\\Tests\\Project1"
"C:\\Temp"
"D:\\"
```

Other non-CWD drive will return root.

1: term “relativization” is from Java.nio.

Path relativization[1]

- In contrast to normalization, path can also be “denormalized” by converting an absolute path to relative path.
  - More generally, given a path `b`, how can it be transformed to path `a` with shortest components.
  - For example, `path{ "/a/d" }.relative("/a/b/c")` would be `"../../d"`.
- Similarly, you can use two ways:
  - Lexical: by `a.lexically_relative(b)`; the process is:
    ① Check whether it’s possible to transform `b` to `a`.
    - If it’s impossible (i.e. conditions below), return empty path directly.

— `root_name() != base.root_name()` is `true`, or — E.g. two paths in different drives in Windows.

— `is_absolute() != base.is_absolute()` is `true`, or — Absolute path + relative path, not lexically transformable.

— `!has_root_directory() && base.has_root_directory()` is `true`, or — E.g. `bar` and `/foo` on Windows, i.e. you cannot cd from `/foo` to `bar` without CWD.

— any *filename* in `relative_path()` or `base.relative_path()` can be interpreted as a *root-name*,

E.g. for UNC path on Windows, e.g. `\\.\C:\Test` uses `C:\Test` as relative path.
That is, UNC doesn’t participate in lexical relative process.

Path relativization

```
a = D:\test\test.txt
b = D:\test\test2\test.txt\..
```

② Determine the first mismatched component of two paths (just like `std::mismatch`).

- Let remaining mismatched components of `a` be $[a_1, a_2)$ and `b` be $[b_1, b_2)$;
  - Example above is $[a_1, a_2) = [test.txt], [b_1, b_2) = [test2, test.txt, ..]$.
- If no mismatched component (i.e. $a_1 = a_2, b_1 = b_2$), return `path{ “.” }`;
- Otherwise, assuming that in $[b_1, b_2)$, n components are `..` and m components are not `..`, `.` and empty.
  - Example above is $n = 1, m = 2$.
  - If $n > m$ (that is, the lexically normalized form contains only `..`), then return empty path.
  - If $n = m$ (that is, the lexically normalized form is `.`), then:
    - If $a_1 = a_2$, return `path{ “.” }`;
    - Otherwise, return $[a_1, a_2)$.
  - If $n < m$, then return a path with 1. `".."` repeated for $m - n$ times; 2. $[a_1, a_2)$.

```
result = ..\test.txt
```

Path relativization

- Actually, this algorithm may falsely report empty path even when such transformation should be possible.
  - For example:

```
fs::path p{ "a/b"};
std::cout << p.lexically_relative("a/b/..") << "\n";
```

$n = 1, m = 0$

```
Program returned: 0
    ""
```

    - Though theoretically it can be "..".
  - If you want an always-correct lexical transformation, you need to do lexical normalization first.

[*Note 3*: If normalization ([fs.path.generic]) is needed to ensure consistent matching of elements, apply `lexically_normal()` to `*this`, `base`, or both. — *end note*]

- The second way is filesystem-dependent `stdfs::relative(a, b)`, which will always ensure correct relative path in the file system.
  - It's same as `lexically_relative` two `weakly_canonical` paths.

Path proximation

- Finally there also exists proximation, which means “relativization if possible, otherwise return original path”.
  - Effectively:

```
path lexically_proximate(const path& b)
{
    if (auto rel = a.lexically_relative(b); !rel.empty())
        return rel;
    return *this;
}
```

  - And `stdfs::proximate(a, b)` is also same as `lexically_proximate` two `weakly_canonical` paths.
  - BTW, when `b` is not provided in `relative` and `proximate`, `current_directory` will be used.

```
friend path operator/( const path& lhs, const path& rhs );
```

```
path& operator/=( const path& p );
```

```
template< class Source >
path& operator/=( const Source& source );
```

```
template< class Source >
path& append( const Source& source );
```

```
template< class InputIt >
path& append( InputIt first, InputIt last );
```

Path composition

- For a given path `./sub/path/file.txt`, it's essentially a combination of hierarchical components.
- C++ provides two utilities to combine components.
  1. Append: combine two components with a directory separator, if needed.
     - For example: (on Linux)

```
std::filesystem::path p = "/home";
//  p == /home/tux/.fonts
std::cout << p / "tux" / ".fonts" << '\n';
```

```
std::filesystem::path p = "/home";
//  p == /home/tux/.fonts
std::cout << p / "tux/" / ".fonts" << '\n';
```

     - However, there exist lots of corner cases…

① Subpath is absolute path: C++ chooses to overwrite (replace) LHS.

- For example:

```
// On Windows,
path("foo") / "C:/bar";  // the result is "C:/bar" (replaces)
```

- But this can be astonishing:

```
std::filesystem::path p = "/home";
//  p == /.fonts
std::cout << p / "tux" / "/.fonts" << '\n';
```

Reason: "`/.fonts`" is absolute path.

Path composition

- DOS-like behavior on Windows also causes surprising result even when subpath is relative.
  - ② No separator is inserted for a single drive; it's still a relative path.

```
std::cout << fs::path{ "C:" } / "Users" / "Admin" << '\n';
std::cout << fs::path{ "C:\\" } / "Users" / "Admin" << '\n';
```

```
"C:Users\\Admin"
"C:\\Users\\Admin"
```

  - ③ Appending a relative path that has different drive will replace the whole path;

```
path("foo") / "C:/bar";  // the result is "C:/bar" (replaces)
path("foo") / "C:";      // the result is "C:"     (replaces)
```

  - ④ Appending a relative path that has same drive will append as if LHS is CWD.

```
std::cout << fs::path{ "C:\\foo" } / "C:bar" << "\n";
std::cout << fs::path{ "C:foo" } / "C:bar" << "\n";
```

```
"C:\\foo\\bar"
"C:foo\\bar"
```

  - ⑤ Appending a relative path that has root directory but no drive, to another path with drive will reserve LHS drive.

```
std::cout << fs::path{ "C:\\bar" } / "\\foo" << "\n";
std::cout << fs::path{ "C:bar" } / "\\foo" << "\n";
```

```
"C:\\foo"
"C:\\foo"
```

Path composition

```
template< class Source >
path& concat( const Source& source );
```
(7)

```
template< class InputIt >
path& concat( InputIt first, InputIt last );
```
(8)

2. Concatenate: combine two components as if concatenating underlying strings directly; no additional separator is introduced.

```
path& operator+=( const path& p );
```
(1)

```
path& operator+=( const string_type& str );
path& operator+=( std::basic_string_view<value_type> str );
```
(2)

```
path& operator+=( const value_type* ptr );
```
(3)

```
path& operator+=( value_type x );
```
(4)

```
template< class CharT >
path& operator+=( CharT x );
```
(5)

```
template< class Source >
path& operator+=( const Source& source );
```
(6)

These overloads are designed to mimic overloads of `std::string::operator+=`.

- Strangely, there is no `operator+`; but normally this operation is used to concatenate with a string, so you can just `operator+` all strings first.
  - Or you have to use either `((stdfs::path{a} += b) += c)`… or `.native()` to use `operator+` of `std::basic_string`.

Path composition

- Note 1: Due to associativity, `p / "a" / "b"` is legal while `p /= "a" /= "b"` is illegal.
  - `p / "a" / "b"` ⇔ `((p / "a") / "b")`, while
  - `p /= "a" /= "b"` ⇔ `(p /= ("a" /= "b"))`.
    - And it's illegal for two string literals to `/=`.
  - You have to add a bunch of brackets; that's why we use `((stdfs::path{a} += b) += c)`….
- Note 2: there also exist some boolean observers to check existence.
  - "empty" means the underlying string contains nothing.
  - And `has_xxx` means whether `xxx` is empty or not.

```
empty

has_root_path
has_root_name
has_root_directory
has_relative_path
has_parent_path
has_filename
has_stem
has_extension

is_absolute
is_relative
```

Path iteration

- As a combination of many different components, path can also be iterated (grouped by separator) in generic format.
- It provides `.begin()` and `.end()` that return a path const iterator.
  - It just iterates through root name, root directory and filenames.
  - For example:

```cpp
#include <filesystem>
#include <iostream>
namespace fs = std::filesystem;

int main()
{
    const fs::path p =
#   ifdef _WIN32
        "C:\\users\\abcdef\\AppData\\Local\\Temp\\";
#   else
        "/home/user/.config/Cppcheck/Cppcheck-GUI.conf";
#   endif
    std::cout << "Examining the path " << p << " through iterators gives\n";
    for (auto it = p.begin(); it != p.end(); ++it)
        std::cout << *it << " | ";
    std::cout << '\n';
}
```

```
--- Windows ---
Examining the path "C:\users\abcdef\AppData\Local\Temp\" through iterators gives
"C:" | "/" | "users" | "abcdef" | "AppData" | "Local" | "Temp" | "" |

--- UNIX ---
Examining the path "/home/user/.config/Cppcheck/Cppcheck-GUI.conf" through iterators gives
"/" | "home" | "user" | ".config" | "Cppcheck" | "Cppcheck-GUI.conf" |
```

Deferenced result (i.e. `iterator::value_type`) is another `path`.

Path iteration

- Though it seems to be bidirectional iterator, it's actually only regulated to be input iterator.
  - Reason: before C++20, forward iterator has such a regulation:
- If `i` and `j` are both dereferenceable, then `i == j` if and only if `*i` and `*j` are bound to the same object.
  - That is, every component should have a fixed source to make every iterator dereferenced to that source.
  - This requires `path` to store a container of components, making it expensive to construct any `path`…
    - And this is how libstdc++ implements it, making it bidirectional.
- However, path iterator is quite like a string `std::views::split` by separator!
  - Another way (as libc++ and MS-STL do) is to cache the range in the iterator, so only when iterator is used will parsing begins.

Issue 2674: Bidirectional iterator requirement on path::iterator is very expensive

Path iteration

- So instead, the standard regulates that:

> 2 A `path::iterator` is a constant iterator meeting all the requirements of a bidirectional iterator except that, for dereferenceable iterators a and b of type `path::iterator` with a `==` b, there is no requirement that `*a` and `*b` are bound to the same object. Its `value_type` is `path`.

  - which makes it only satisfies input iterator, and thus it's impossible to apply some functions in `<algorithm>`.

- BUT, such requirement is not part of `bidirectional_iterator` in C++20!
  - So theoretically it should be able to utilize constrained algorithms, i.e. `std::ranges::xxx`.
  - Well, this problem is much more complicated… See our homework for discussion.
    - Anyway, libc++ already adds `iterator_concept` for it, while MS-STL cannot do so.

Path modification

Return reference to `*this`.

Modifiers

- `clear`
- `make_preferred`
- `remove_filename`
- `replace_filename`
- `replace_extension`
- `swap`

- There also exist some simple non-const methods:
  1. `.make_preferred()`: for path whose native format is also generic format, convert current separators to preferred separators.
     - For example:

```
fs::path p{ "C:/Test\\Test2" };
std::cout << p << "\n" << p.make_preferred() << "\n";
```

```
"C:/Test\\Test2"
"C:\\Test\\Test2"
```

  2. `.remove_filename()`: Remove the last component (if it exists) so `.has_filename()` returns `false`.
     - So after removal, the path is either empty or ends with a separator.

```
std::cout << std::boolalpha
          << (p = "foo/bar").remove_filename()
          << (p = "foo/").remove_filename() <<
          << (p = "/foo").remove_filename() <<
          << (p = "/").remove_filename() << '\
          << (p = "").remove_filename() << '\t
```

```
"foo/"
"foo/"
"/"
"/"
""
```

"foo" will also be converted to "".

Path modification

3. `.replace_filename(const path& rep)`: equivalent to 1. `this->remove_filename()`; 2. `(*this) /= rep`.
4. `.replace_extension(const path& rep = {})`: equivalent to code below:
   - For example:

Path:	Ext:	Result:
"/foo/bar.jpg"	".png"	"/foo/bar.png"
"/foo/bar.jpg"	"png"	"/foo/bar.png"
"/foo/bar.jpg"	"."	"/foo/bar."
"/foo/bar.jpg"	""	"/foo/bar"
"/foo/bar."	"png"	"/foo/bar.png"
"/foo/bar"	".png"	"/foo/bar.png"
"/foo/bar"	"png"	"/foo/bar.png"
"/foo/bar"	"."	"/foo/bar."
"/foo/bar"	""	"/foo/bar"
"/foo/."	".png"	"/foo/..png"
"/foo/."	"png"	"/foo/..png"
"/foo/."	"."	"/foo/.."
"/foo/."	""	"/foo/."
"/foo/"	".png"	"/foo/.png"
"/foo/"	"png"	"/foo/.png"

```
// Assume we have a function TryRemoveExtension,
// which will remove extension if it exists.
path& ReplaceExtension(const path& rep = {})
{
    TryRemoveExtension();
    if (rep.empty)
        return *this;
    // Add '.' if necessary
    if (rep.native()[0] != DOT)
        this->concat(DOT);
    return *this += rep;
}
```

Path

1) If `root_name().native().compare(p.root_name().native())` is nonzero, returns that value.

Otherwise, if `has_root_directory() != p.has_root_directory()`, returns a value less than zero if `has_root_directory()` is `false` and a value greater than zero otherwise.

Before element-wise comparison, it needs to judge these conditions first.

- And finally some simple utilities, just list here.
  - Comparable (by `<=>/==` or `.compare`); compare **components**. e.g. `"D:/Test" == "D://Test"`
  - Hashable (by `std::hash` or `friend hash_value`); hash components.
  - Input / Output by >> / <<;
    - For `i/ostream<CharT, Traits>`, equiv. to input / output the `.string<CharT, Traits>()` with `std::quoted` so space will not interrupt input.

```
"C:\\foo\\bar"
"C:foo\\bar"
```

    - Recap: `quoted` will escape the quote and the escape, so `\` is escaped to `\\`.
  - Formattable **since C++26**.

Format specification

The syntax of format specifications *path-format-spec* is:

fill-and-align(optional) *width*(optional) **?**(optional) **g**(optional)

fill-and-align and *width* have the same meaning as in standard format specification.

The **?** option is used to format the pathname as an escaped string.

The **g** option is used to specify that the pathname is in generic-format representation.

Path formatting

- Its regulation is slightly different from `.string()`.

std::formatter<std::filesystem::path>::format

```
template< class FormatContext >
auto format( const std::filesystem::path& p, FormatContext& ctx ) const
    -> FormatContext::iterator;
```

Let `s` be `p.generic_string<std::filesystem::path::value_type>()` if the **g** option is used, otherwise `p.native()`. Writes `s` into `ctx.out()` as specified by *path-format-spec*.

For character transcoding of the pathname:

- The pathname is transcoded from the native encoding for wide character strings to UTF-8 with maximal subparts of ill-formed subsequences substituted with U+FFFD REPLACEMENT CHARACTER if
  - `std::is_same_v<CharT, char>` is `true`,
  - `std::is_same_v<typename path::value_type, wchar_t>` is `true`, and
  - ordinary literal encoding is UTF-8.
- Otherwise, no transcoding is performed if `std::is_same_v<typename path::value_type, CharT>` is `true`.
- Otherwise, transcoding is implementation-defined.

Returns an iterator past the end of the output range.

i.e. encoding of char string literal, as specified in compiler execution charset option.

i.e. explicitly regulate that in UTF-8 execution charset on Windows, illegal character will be converted to U+FFFD.

As a C++26 feature, it's not yet implemented so "implementation-defined" is unknown (but likely to be printable with `std::print`).

Final Notes

- Note 1: there also exists `.u8path()` to construct from a UTF-8 string in C++17, which is then deprecated in C++20.
  - Reason: C++20 introduces `char8_t`, which distinguishes UTF-8 sequence from `char` by template so there is no need to introduce a new method.
  - For the same reason, `.(generic_)u8string` returns `std::string` in C++17 and `std::u8string` in C++20.
- Note 2: to use `path` as key in map, you usually need to normalize it first by `canonical`.
  - Reason: comparison and hashing of `path` are performed **lexically** for the underlying components.
  - Without normalization, two equivalent paths may be seen as two keys.
  - On Windows, you may even need to `to_lower` all case-insensitive paths.
  - A more expensive but always-correct way is by `stdfs::equivalent` to compare, which needs file system call to check equality of two paths. Covered later.

Supplementary

- File system
  - Path operations
  - File system operations
    - Overview
    - File status query and directory iteration
    - Modification operations

Overview

- In this section, most of the functions interact with the underlying filesystem, which are in the global scope `stdfs::`.
  - By contrast, functions we taught in the last section are purely lexical (and thus cheaper), which are member functions of `stdfs::path`.
- Almost every function provides two versions:
  1. Error code version: add `std::error_code&` as the last parameter to return filesystem error (`noexcept` if only filesystem error is possible);
  2. Exception version: throw `stdfs::filesystem_error` to represent error.

  - Reason: filesystem operations can easily incur TOC/TOU problem, so pre-check cannot prevent error.
    - Thus, it's hard to predict whether error occurrence is in hot path or not, making exceptions sometimes not proper.

Overview

- For example, if we want to write a “chmod when it exists”.
  - So pseudocode can be:

```
function MyCHMOD(path) -> bool:
    if not exists(path) then return false
    chmod(path, WRITE_PERMISSION)
    return true
end
```

- But filesystem is **cross-process**, so the events can be:
  - We check that file indeed exists (TOC);
  - Another process removes this file;
  - We change permission of the file, which doesn’t exist and causes error (TOU).
- When multiple processes access the same filesystem object (i.e. *race*), the specific behavior is implementation-defined.

- Even worse, attackers can easily leverage TOC/TOU.
  - A well-known one is `stdfs::remove_all` (also in Rust!), which removes all files under the directory recursively but skip removing files in "symbolic links".
    - We'll cover symbolic links later; but generally it means an object that refers to another directory.
    - Such skip can prevent users from accidentally removing files in other folders.
  - And all three standard libraries initially implement it as:
    1. Check whether the object is a symbolic link;
    2. If it is not, recursively remove its files.
  - Say hackers want to remove `sensitive/` but they don't have permission; but current system runs a privileged program with `stdfs::remove_all`, which periodically removes `recyclebin/` that don't add permission.
    1. Hackers create a directory called `temp` in `recyclebin/` first;
    2. `remove_all` checks that it's not a symbolic link;
    3. Hackers delete `temp` and create a symbolic link `temp` to `sensitive/`.
    4. `remove_all` removes all files in `temp`, i.e. all files in `sensitive`. Hackers win!

Overview

- Similarly, user code can become vulnerable for TOC/TOU…
- Core problem: “path name” is a mutable property; a more robust way to always refer to the same filesystem object should be some handle.
  - Which is proposed in P1883 (i.e. the low-level file i/o library);
  - However, this can make filesystem APIs very obscure to use.

-------- Back to standard ---------

- APIs of `stdfs::filesystem_error` are quite simple:
  - We don’t dig into `std::error_code` here; see our homework for details.

Member functions

(constructor)	constructs the exception object (public member function)
`operator=`	replaces the exception object (public member function)
`path1` `path2`	returns the paths that were involved in the operation that caused the error (public member function)
`what`	returns the explanatory string (public member function)

Inherited from std::system_error

Member functions

`code`	returns error code (public member function of `std::system_error`)
`what` [virtual]	returns an explanatory string (virtual public member function of `std::system_error`)

Supplementary

- File system
  - Path operations
  - File system operations
    - Overview
    - File status query and directory iteration
    - Modification operations

File status

- A file has the following attributes:
  - Name, as used in path;
  - Type;
  - Permission;
  - Size;
  - Last modification time.

  C++ uses POSIX conventions, but allows slight customization for different systems (e.g. Windows).
- And POSIX regulates the following file types (as reflected in `stdfs::file_type`):
  - Implementations are allowed to add new types (e.g. `junction` in MS-STL).

```
enum class file_type {
    none = /* unspecified */,
    not_found = /* unspecified */,
    regular = /* unspecified */,
    directory = /* unspecified */,
    symlink = /* unspecified */,
    block = /* unspecified */,
    character = /* unspecified */,
    fifo = /* unspecified */,
    socket = /* unspecified */,
    unknown = /* unspecified */,
    /* implementation-defined */
};
```

File types

Enumerator	Meaning
`none`	indicates that the file status has not been evaluated yet, or an error occurred when evaluating it
`not_found`	indicates that the file was not found (this is not considered an error)
`regular`	a regular file
`directory`	a directory
`symlink`	a symbolic link
`block`	a block special file
`character`	a character special file
`fifo`	a FIFO (also known as pipe) file
`socket`	a socket file
`unknown`	the file exists but its type could not be determined

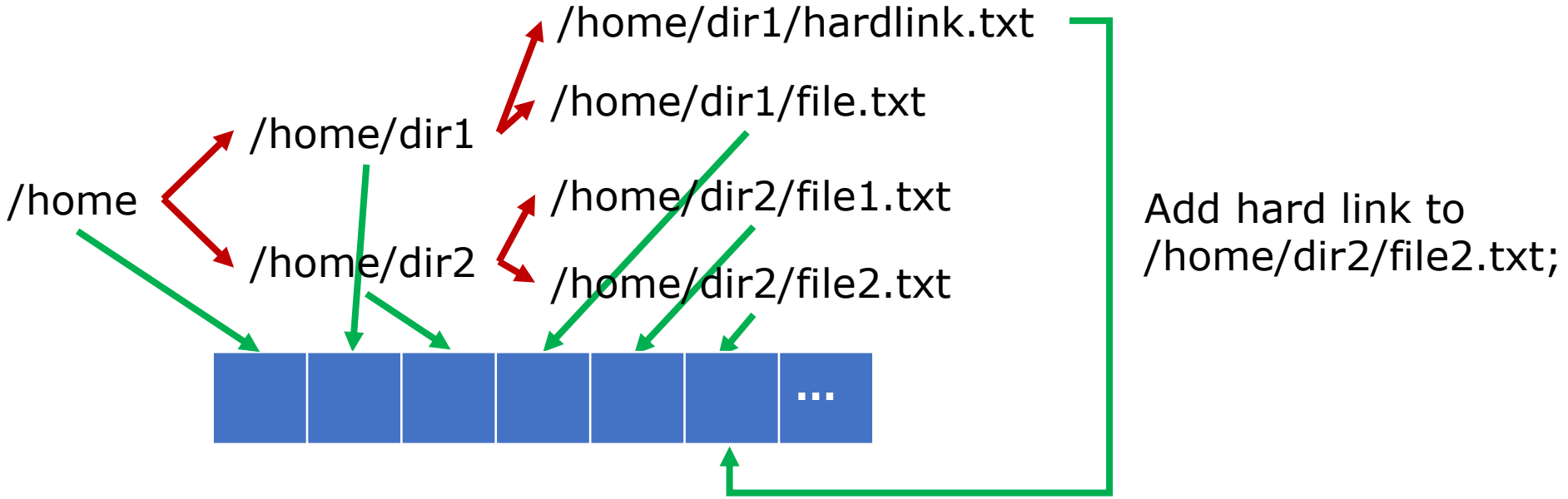
What are these types?

Link

- There are many kinds of links in the filesystem.
  - Essentially, links are just objects that *redirect* to other objects;
  - Different links are just redirection at different levels.
- A simple illusion of how filesystem handles objects:

Hard Link

1. Hard link: as if adding reference count to underlying data nodes.
   - Filesystem cannot distinguish a hard link with the original object (so there is no type called “hard link”!).
   - When you delete `/home/dir2/file2.txt`, the content isn’t really deleted; `/home/dir1/hardlink.txt` still keeps the node.
   - Hard link count of an object can be checked by `stdfs::hard_link_count(p)`.



Hard Link

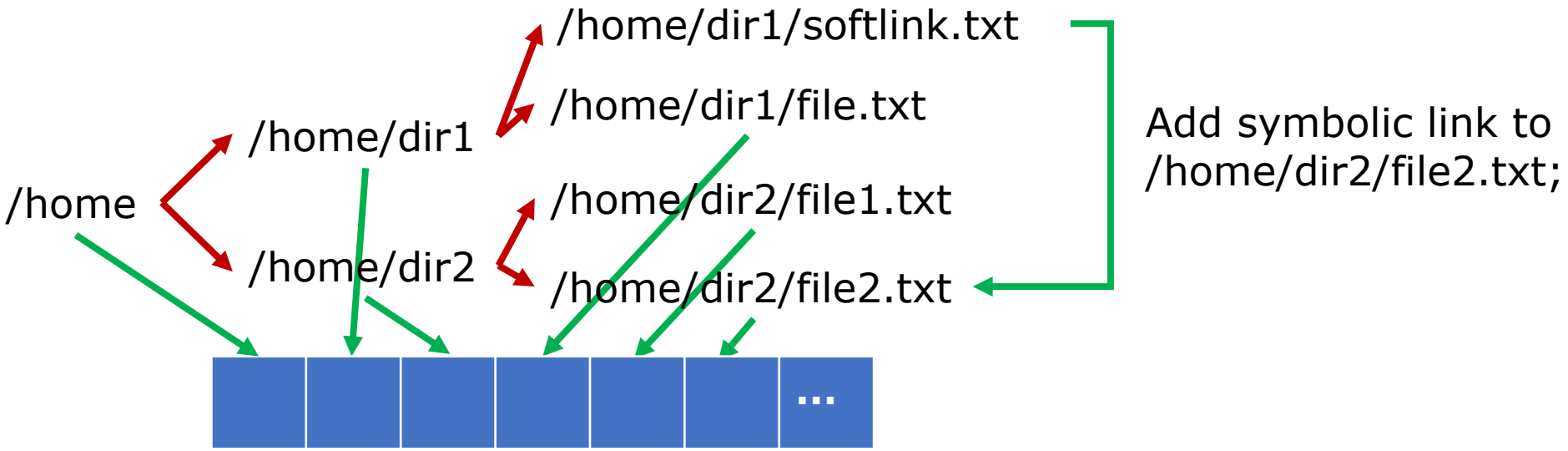
- Most of the filesystems don't allow users to create hard link to directories easily.
  - Core reason: filesystem is usually assumed to form a **tree**.
    - However, creating hard link of some directory in its descendent will make a cycle in the graph, making it not a tree.
    - Some system programs may traverse the filesystem without special check; a circle will cause infinite loop.
  - Linux: `ln hd.txt a.txt`; do not allow hard link to directory.
  - MacOS: `ln hd.txt a.txt`; no native command for directory but can use POSIX `link()`;
    - MacOS `link` implementation checks whether new hard link to directory causes cycle; allow if not (but possibly need root privilege).
  - Windows: `mklink /H hd.txt a.txt`; do not allow hard link to directory.

Hard Link

- Note 1: `hard_link_count` for directory is implementation-defined.
  - Unix: "." and ".." are all seen as hard link; so a new directory will have `hard_link_count` as `2` and `++hard_link_count` of its parent directory.
  - Windows: always return `1`.
  - Return type: `uintmax_t`. The non-throwing overload returns `static_cast<uintmax_t>(-1)` on errors.
- Note 2: hard link has some other restrictions:
  1. Some filesystems don't support hard link (notably FAT file system; most of USB flash drives (U盘) use it).
  2. Hard link is not cross-filesystem since different filesystems may have different data structures. Hard link exists only in the same filesystem.
     - Particularly on Windows, they must be in the same volume (notably drive).
  3. Some filesystems may have limits for hard links per file.

Symbolic Link

2. Symbolic link (soft link): as if pointing to filesystem entry.
   - Or, you can think it as a `weak_ptr` to the underlying node.
   - When the real entry `/home/dir2/file2.txt` is deleted, its content will also be deleted though symbolic link `/home/dir1/softlink.txt` exists.
     - Any redirection operation of the soft link later (e.g. file I/O) will fail.



Filesystem functionality can be checked in MS Doc.
Notice that UDF is widely used in optical discs (光盘).

Symbolic Link

- Filesystem treats it as a distinct file type, but most native APIs will automatically redirect it to the real entry.
  - That is, users (instead of filesystem!) cannot distinguish symbolic link with a normal object, unless they use a few special functions.
- Therefore, filesystem allows symbolic link to directory.
- Unix: `ln -s sym a`;
- Windows: `mklink /D sym a`; **Particularly**, creating symbolic link on Windows requires root privilege.

- Note: Some filesystems don't support soft link (notably FAT).
  - But as it points to filesystem entry, it's cross-filesystem.
  - Therefore, it's okay to create soft link to FAT on NTFS since NTFS supports it (but not vice versa).

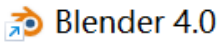
Junction

3. For non-root privilege, Windows allows link to directory by a new type called “junction”.
   - `mklink /J junc a`.
   - Like symbolic link, when the real node is deleted, junction still exists but becomes invalid.
   - Though junction is cross-filesystem, it only allows to link directory in local computer.
     - i.e. no support for network path and UNC.

- Most of `stdfs` APIs will follow all links above and resolve to the final real entry.
  - For example, `stdfs::exists`; `stdfs::absolute`; and thus `stdfs::canonical`; `stdfs::relative`; `stdfs::proximate`.

Check this blog for link details on Windows.

Shortcut



- BTW, there also exists “shortcut” (快捷方式) on Windows, which is actually a “user-space symbolic link”.
  - It’s a file with “.lnk” extension; filesystem just treats it as a normal file (and it is!).
  - Its content is the real path of the filesystem object, how it should be launched, etc.
  - So guess: when you click shortcut, how are you redirected to the real entry?
    - Yes, Windows Explorer (文件资源管理器) UI helps you!
    - The Explorer program automatically redirects your click as if you click the real entry; other programs that don’t do special process **cannot redirect**.
    - By contrast, links will be redirected by filesystem automatically even if programs don’t do special process.

Character and Block File

- When you plug in a USB (mouse, disk, etc.), how can OS identify the way to interact with the connected device?
  - By **drive** program, if you've learnt *Embedded System*.
  - When you want to support a new external device, you can write your own drive with a bunch of functions (e.g. `open`, `write`, etc.) to make OS know how to interact with it.
  - And finally, your drive needs to expose a *device file* to make users able to `open`, `write`, etc. by OS APIs.
- Character file and block file are just such device files.
  - When you read from / write to them, it's as if you get data from / input data to your drive and thus successfully interact with your device.
  - Character file means "I/O is passed directly without buffering"; e.g. terminal: `/dev/tty`
  - Block file means "Buffering I/O and pass until proper time". e.g. SSD: `/dev/sda`

Windows also supports character file (e.g. CON) but MS-STL always returns `false` for these two types since it's expensive to detect such file type.

FIFO

- FIFO is just named pipe.
  - In Linux, we can pass output of a program as input to another program by `|` (e.g. `cat data.txt | grep "info"`).
    - Pipe allows the data to keep in memory instead of writing into real files (and thus real storage) for inter-process communication.
  - Sometimes, it's inconvenient to create anonymous pipes and pass them to other programs.
    - Instead, you can create a named pipe in the filesystem, with some programs writing and other programs reading it (again, happen in memory without really going down to real storage).
    - E.g. in command line: `mkfifo test; cat data.txt > test &; grep "info" test`; or by `mkfifo` in POSIX API.
  - So `test` just has fifo type, which is specially interpreted in POSIX system.

Socket

```
// We omit error code handling here.
sockaddr_in sockAddr{};
sockAddr.sin_family = AF_INET;
::inet_pton(AF_INET, argv[1], &sockAddr.sin_addr);
sockAddr.sin_port = ::htons(port);
listenfd = ::socket(AF_INET, SOCK_STREAM, 0);
::bind(listenfd, (sockaddr*)(&sockAddr), sizeof(sockaddr_in));
// For server: listen and accept
// For client: connect by sockAddr
```

Fill network location.

- Similarly, sometimes it's inconvenient to use network socket to communicate across processes.
  - Instead, you can create a socket file (a.k.a. Unix domain socket, UDS).
  - We know how to create a network server socket in Linux.
  - For UDS, just change flag and location to local socket file:

```
sockaddr_un sockAddr;
sockAddr.sun_family = AF_UNIX;
// Specify file path, less than 108 bytes in Linux.
std::strcpy(sockAddr.sun_path, "sockfile.sock");
listenfd = ::socket(AF_UNIX, SOCK_STREAM, 0);
::bind(listenfd, (sockaddr*)(&sockAddr), sizeof(sockaddr_un));
// For server: listen and accept
// For client: connect by sockAddr
```

After `bind`, process will create `sockfile.sock` in CWD, which has file type as `socket`.

  - However, `bind` requires the address to be not already used, so the socket file shouldn't exist before `bind` (thus, it's usually a temporary file).

FIFO and Socket

- Note 1: difference between fifo and socket:
  1. Socket file can use `send` or `recv`, as network socket does; fifo can only use `write` or `read`.
  2. Socket can use datagram instead of byte stream (by `SOCK_DGRAM`), while fifo only uses byte stream.
  3. Socket is bidirectional, i.e. server and client can send or recv the other freely; fifo is unidirectional, i.e. sender always send and receiver always recv.
  4. Typically, fifo is used for two users, while socket can be used for multiple clients.
- Note 2: Windows also supports fifo (since Windows 2000 professional) and UDS (since Windows 10 17063 (2017/12)).
  - However, MS-STL always returns `false` for `is_fifo` and `is_socket`.

```
_EXPORT_STD _NODISCARD inline bool is_socket(const path&) noexcept /* strengthened */ {
    // tests whether the input path is a socket (never on Windows)
    return false;
}
```

Reason: UDS doesn't create a real file like in UNIX.

Permission

- For permission, there are three levels in POSIX: user / group / all.
  - Each level can have read / write / execution right.
  - In Linux bash, we can change permission by `chmod`; e.g. chmod 764 means that owner can read, write or execute, current group can read and write, while all others can only read.
- As each level can be expressed in three bits, you can use octal number to form the permission in C++.
  - E.g. literal 0764, since leading 0 means octal literal.
- However, Windows permission system (Active Control List, ACL) is not compatible with POSIX. It thus uses naïve DOS permission:
  1. All users can read, write and execute the file (777);
  2. All users can read and execute the file (555).

Permission

- In POSIX (again, not in Windows), there also exist three special permissions: `setuid`, `setgid` and `sticky bit`.
  - `setuid`: when executing the binary, use the owner's user id (so that it can execute code that only owner can execute).
    - Typical example: `/bin/passwd`, which will write `/etc/shadow`. Users can change password by `/bin/passwd`, but don't have privilege to write `/etc/shadow` (otherwise they can change anyone's password!).
      - Thus, `/bin/passwd` has permission `setuid`, i.e. it has root privilege to write `/etc/shadow` when executing `/bin/passwd`.
      - Since `/bin/passwd` controls what it accesses, it's still safe.
  - `setgid`: when executing the binary, use the owner's group id.
    - Typical example: `/usr/bin/wall`, which needs group privilege to write to others' `tty`.

Permission

- `sticky bit`: typically means "when applying to a directory, only file owner can delete his/her file even if he/she can write the directory".
  - Typical example: `/tmp`; all users have write permission to `/tmp` to keep temporary files. They may delete their cache files later.
    - However, this then allows the user to delete others' file, which may randomly crash others' applications.
  - Solution 1: every user protects his/her file in `/tmp` by e.g. `chmod 007`.
    - Well, that's too troublesome…
  - Solution 2: apply sticky bit to `/tmp`!
    - Then only file owner can delete his/her file.

- C++ defines these permission specifications with enumeration.

Permission

- Permission values are defined as `enum class perms` with these enumerators:
  - With overloaded bit operators.

Member constants

Member constant	Value (octal)	POSIX equivalent	Meaning
`none`	`0`		No permission bits are set
`owner_read`	`0400`	`S_IRUSR`	File owner has read permission
`owner_write`	`0200`	`S_IWUSR`	File owner has write permission
`owner_exec`	`0100`	`S_IXUSR`	File owner has execute/search permission
`owner_all`	`0700`	`S_IRWXU`	File owner has read, write, and execute/search permissions Equivalent to `owner_read | owner_write | owner_exec`
`group_read`	`040`	`S_IRGRP`	The file's user group has read permission
`group_write`	`020`	`S_IWGRP`	The file's user group has write permission
`group_exec`	`010`	`S_IXGRP`	The file's user group has execute/search permission
`group_all`	`070`	`S_IRWXG`	The file's user group has read, write, and execute/search permissions Equivalent to `group_read | group_write | group_exec`
`others_read`	`04`	`S_IROTH`	Other users have read permission
`others_write`	`02`	`S_IWOTH`	Other users have write permission
`others_exec`	`01`	`S_IXOTH`	Other users have execute/search permission
`others_all`	`07`	`S_IRWXO`	Other users have read, write, and execute/search permissions Equivalent to `others_read | others_write | others_exec`
`all`	`0777`		All users have read, write, and execute/search permissions Equivalent to `owner_all | group_all | others_all`
`set_uid`	`04000`	`S_ISUID`	Set user ID to file owner user ID on execution
`set_gid`	`02000`	`S_ISGID`	Set group ID to file's user group ID on execution
`sticky_bit`	`01000`	`S_ISVTX`	Implementation-defined meaning, but POSIX XSI specifies that when set on a directory, only file owners may delete files even if the directory is writeable to others (used with `/tmp`)
`mask`	`07777`		All valid permission bits. Equivalent to `all | set_uid | set_gid | sticky_bit`

Additionally, the following constants of this type are defined, which do not represent permissions:

Member constant	Value (hex)	Meaning
`unknown`	`0xFFFF`	Unknown permissions (e.g. when `filesystem::file_status` is created without permissions)

`perms` satisfies the requirements of *BitmaskType* (which means the bitwise operators `operator&`, `operator|`, `operator^`, `operator~`, `operator&=`, `operator|=`, and `operator^=` are defined for this type). `none` represents the empty bitmask; every other enumerator represents a distinct bitmask element.

File status query

- To test file type, you can use these APIs:
  - Params: `(const path& p[, error_code])`.

`is_block_file` (C++17)	checks whether the given path refers to block device (function)
`is_character_file` (C++17)	checks whether the given path refers to a character device (function)
`is_directory` (C++17)	checks whether the given path refers to a directory (function)
`is_fifo` (C++17)	checks whether the given path refers to a named pipe (function)
`is_other` (C++17)	checks whether the argument refers to an *other* file (function)
`is_regular_file` (C++17)	checks whether the argument refers to a regular file (function)
`is_socket` (C++17)	checks whether the argument refers to a named IPC socket (function)
`is_symlink` (C++17)	checks whether the argument refers to a symbolic link (function)
`status_known` (C++17)	checks whether file status is known (function)

Other file: file exists but isn't regular file / directory / symlink.

File status query

- However, a more efficient way is to query type once and cache it, so every test will be a cheap and plain integer comparison.
  - Just by `stdfs::status`!
  - For example:

```
switch (fs::path p{argv[1]}; status(p).type()) {
  case fs::file_type::not_found:
    std::cout << "path \"" << p.string() << "\" does not exist\n";
    break;
  case fs::file_type::regular:
    std::cout << '"' << p.string() << "\" exists with "
              << file_size(p) << " bytes\n";
    break;
  case fs::file_type::directory:
    std::cout << '"' << p.string() << "\" is a directory containing:\n";
    for (const auto& e : std::filesystem::directory_iterator{p}) {
      std::cout << "  " << e.path().string() << '\n';
    }
    break;
  default:
    std::cout << '"' << p.string() << "\" is a special file\n";
    break;
}
```

Credit: C++17 the Complete Guide, *Nicolai M. Josuttis*.

File status query

- `status` essentially returns `file_status`, with `.type()->file_type` and `.permissions()->perms`:
  - Another example:

```
std::string asString(const std::filesystem::perms& pm)
{
  using perms = std::filesystem::perms;
  std::string s;
  s.resize(9);
  s[0] = (pm & perms::owner_read)   != perms::none ? 'r' : '-';
  s[1] = (pm & perms::owner_write)  != perms::none ? 'w' : '-';
  s[2] = (pm & perms::owner_exec)   != perms::none ? 'x' : '-';
  s[3] = (pm & perms::group_read)   != perms::none ? 'r' : '-';
  s[4] = (pm & perms::group_write)  != perms::none ? 'w' : '-';
  s[5] = (pm & perms::group_exec)   != perms::none ? 'x' : '-';
  s[6] = (pm & perms::others_read)  != perms::none ? 'r' : '-';
  s[7] = (pm & perms::others_write) != perms::none ? 'w' : '-';
  s[8] = (pm & perms::others_exec)  != perms::none ? 'x' : '-';
  return s;
}
```

```
std::cout << "permissions: " << asString(status(mypath).permissions())
          << '\n';
```

Member functions

(constructor)	constructs a file_status object (public member function)
operator=	assigns contents (public member function)
(destructor)	implicit destructor (public member function)
type	gets or sets the type of the file (public member function)
permissions	gets or sets the permissions of the file (public member function)

Non-member functions

operator== (C++20)	compares two file_status objects (function)

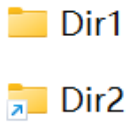
Note: `is_xxx` (e.g. `is_directory`) can also accept `file_status`, which is `noexcept` since it's just plain integer comparison.

File status query

- However, when you try to apply `status` to symbolic links:

```
stdfs::path p{ R"(D:\Work\C++\ExpDir\Dir2)" };
auto status = stdfs::status(p);
std::cout << stdfs::is_directory(status) << " "
          << stdfs::is_symlink(status) << "\n";
```

```
1 0
```



  - Oops, the type is not `symlink`, but directory!
  - Reason: filesystem will automatically follow symbolic links in its APIs.
- To query the shallow status instead of following links, you should use `symlink_status`:

```
stdfs::path p{ R"(D:\Work\C++\ExpDir\Dir2)" };
auto status = stdfs::symlink_status(p);
std::cout << stdfs::is_directory(status) << " "
          << (status.type() == stdfs::file_type::junction) << "\n";
```

```
0 1
```

Note 1: `is_symlink(p[, ec])` internally uses `symlink_status`.

Note 2, again: hard link cannot be distinguished.

Here we create `Dir2` as junction, which is listed as a new enumerator in MS-STL so **`is_symlink` is still `false`** (well, though it can be queried by `symlink_status`).

File status query

MS-STL: return 0;
libc++ & libstdc++: throw error ("is a directory");

- The last two status are size and last modification time, which can also be queried by path:

  The non-throwing overload returns `static_cast<std::uintmax_t>(-1)` on errors.

  - `stdfs::file_size(p[, ec]) -> std::uintmax_t;`
    - The result of directory is implementation-defined.
  - `stdfs::last_write_time(p[, ec]) -> stdfs::file_time_type;`
    - The specific utilities in `std::chrono` will be introduced later.
- Note: except for type, you can change any status directly.
  - `stdfs::resize_file(p, newSize[, ec]);`
    - If `newSize` is greater than current size, fill new space with 0; otherwise truncate the file.
    - Note: on filesystems that support sparse files, greater size of file may not decrease available storage in filesystem as long as "new space with 0" is not really written.

File status query

- `stdfs::last_write_time(p, newTime[, ec]);`
  - But the grain of filesystem may be not same as program time, so there can be rounding error.
- `stdfs::permissions(p, newPerm[, opt[, ec]]);`
  - `opt` is specified as `enum class stdfs::perm_options`, also with bit operators.

At most one of add, remove, replace may be present, otherwise the behavior of the permissions function is undefined.

Enumerator	Meaning
replace	permissions will be completely replaced by the argument to `permissions()` (default behavior)
add	permissions will be replaced by the bitwise OR of the argument and the current permissions
remove	permissions will be replaced by the bitwise AND of the negated argument and current permissions
nofollow	permissions will be changed on the symlink itself, rather than on the file it resolves to

  - Default `opt` is `replace`.

Directory entry

- Umm, so is there a way to query all attributes once and just get them from cache?
  - By `stdfs::directory_entry`!
  - For example:

```cpp
auto entry = stdfs::directory_entry{ R"(D:\Work\C++\ExpDir\Dir2)" };
{
    std::cout << entry
              << " Last modification time = " << entry.last_write_time()
              << ", Hard link count = " << entry.hard_link_count() << "\n";

    if (entry.is_regular_file())
        std::cout << "File size = " << entry.file_size();
    else
    {
        std::cout << "Directory; is symlink : " << entry.is_symlink()
                  << " is junction: "
                  << (entry.symlink_status().type() ==
                      stdfs::file_type::junction);
    }
    std::cout << "\n----------" << std::endl;
```

Construct from path, and ctor will query all attributes if they exist.

Note: in libstdc++, you need to `#include<chrono>` to make time outputable (C++20, covered later).

```
"D:\\Work\\C++\\ExpDir\\Dir2" Last modification time = 2026-02-04 10:29:15.7225417, Hard link count = 1
Directory; is symlink : 0 is junction: 1
----------
```

Directory entry

- It has almost all query methods we covered before:
  - And some utilities (**only** applied to the underlying path!):
    - Comparable
    - `operator<<`

`.is_symlink` uses `symlink_status` while others use `status`. So `.is_symlink && .is_directory` can be `true`.

Observers

`path` `operator const path&`	returns the path the entry refers to (public member function)
`exists`	checks whether directory entry refers to existing file system object (public member function)
`is_block_file`	checks whether the directory entry refers to block device (public member function)
`is_character_file`	checks whether the directory entry refers to a character device (public member function)
`is_directory`	checks whether the directory entry refers to a directory (public member function)
`is_fifo`	checks whether the directory entry refers to a named pipe (public member function)
`is_other`	checks whether the directory entry refers to an *other* file (public member function)
`is_regular_file`	checks whether the directory entry refers to a regular file (public member function)
`is_socket`	checks whether the directory entry refers to a named IPC socket (public member function)
`is_symlink`	checks whether the directory entry refers to a symbolic link (public member function)
`file_size`	returns the size of the file to which the directory entry refers (public member function)
`hard_link_count`	returns the number of hard links referring to the file to which the directory entry refers (public member function)
`last_write_time`	gets the time of the last data modification of the file to which the directory entry refers (public member function)
`status` `symlink_status`	status of the file designated by this directory entry; status of the file/symlink designated by this directory entry (public member function)

Directory entry

- And also some simple modifiers:

`operator=`	assigns contents (public member function)
`assign`	assigns contents (public member function)
`replace_filename`	sets the filename (public member function)
`refresh`	updates the cached file attributes (public member function)

i.e. `path.replace_filename(…)` and `.refresh()`.

i.e. Re-query attributes by filesystem calls.

  - Applying these methods doesn't e.g. rename the underlying file in the filesystem, but just re-query!
- Actually, `directory_entry` is value type of `directory_iterator`, which can be used to iterate over a path.

Directory iteration

1. Besides `operator++`, the iterator also defines `.increment(ec)` to accept error code.
2. Essentially, `begin` returns its own copy and `end` returns default constructed iterator.

- It has `friend begin` and `friend end` and naturally forms a range, making it iterable directly:

```
for (const auto& entry :
    stdfs::directory_iterator{ R"(D:\Work\C++\ExpDir)" })
```

Path will include root (passed in ctor) as prefix.

```
"D:\\Work\\C++\\ExpDir\\Dir1" Last modification time = 2026-02-04 10:29:15.7225417, Hard link count = 1
Directory; is symlink : 0 is junction: 0
----------
"D:\\Work\\C++\\ExpDir\\Dir2" Last modification time = 2026-02-04 10:29:15.7225417, Hard link count = 1
Directory; is symlink : 0 is junction: 1
----------
"D:\\Work\\C++\\ExpDir\\test.txt" Last modification time = 2026-02-05 09:34:49.0577604, Hard link count = 1
File size = 5
----------
```

- You can also add `stdfs::directory_options` as the second param in ctor to control iterator behavior:

Constants

Again, it's bit-operatable scoped enumeration.

Enumerator	Meaning
none	(default) skip directory symlinks, "permission denied" is error
follow_directory_symlink	follow rather than skip directory symlinks
skip_permission_denied	skip directories that would otherwise result in "permission denied" errors

Directory iteration

- To iterate recursively in the directory, you can also use `recursive_directory_iterator`.
  - which will then use implementation-defined way (usually DFS) to traverse all entries.
  - For example:

```
for (const auto &entry : stdfs::recursive_directory_iterator{
        R"(D:\Work\C++\ExpDir)",
        stdfs::directory_options::follow_directory_symlink })
```

```
"D:\\Work\\C++\\ExpDir\\Dir1" Last modification time = 2026-02-04 10:29:15.7225417, Hard link count = 1
Directory; is symlink : 0 is junction: 0
----------
"D:\\Work\\C++\\ExpDir\\Dir1\\test2.txt" Last modification time = 2026-02-04 10:29:14.7068836, Hard link count = 1
File size = 0
----------
"D:\\Work\\C++\\ExpDir\\Dir2" Last modification time = 2026-02-04 10:29:15.7225417, Hard link count = 1
Directory; is symlink : 0 is junction: 1
----------
"D:\\Work\\C++\\ExpDir\\Dir2\\test2.txt" Last modification time = 2026-02-04 10:29:14.7068836, Hard link count = 1
File size = 0
----------
"D:\\Work\\C++\\ExpDir\\test.txt" Last modification time = 2026-02-05 09:34:49.0577604, Hard link count = 1
File size = 5
----------
```

`Dir2` is junction to `Dir1`, and we follow symlink, so essentially contents in `Dir1` is printed again.

Directory iteration

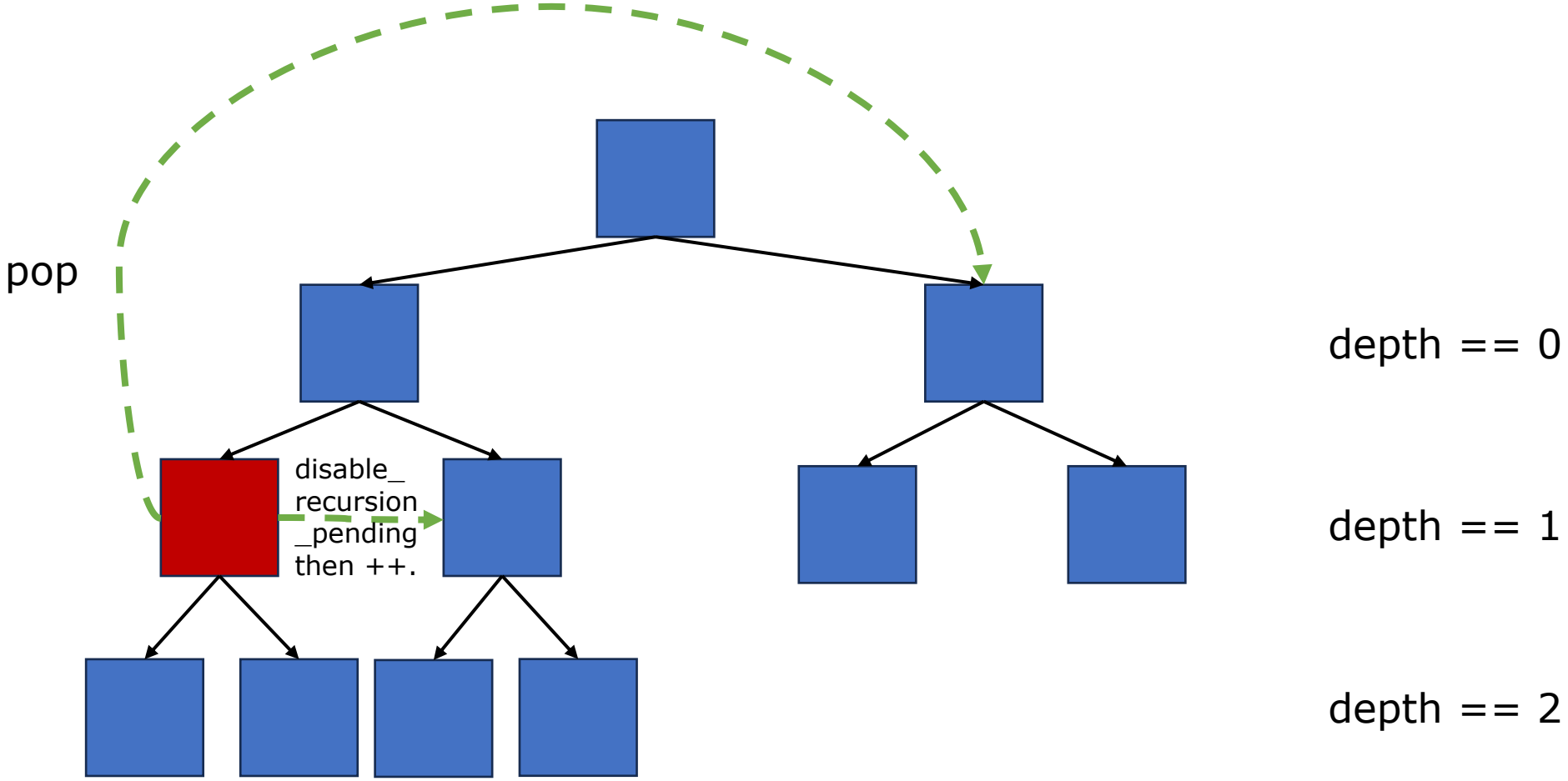
- The iterator has more member methods:

Observer:	`options->directory_options`	returns the currently active options that affect the iteration (public member function)
	`depth->int`	returns the current recursion depth (public member function) (Starting from 0)
	`recursion_pending->bool`	checks whether the recursion is disabled for the current directory (public member function)
Modifier:	`pop`	moves the iterator one level up in the directory hierarchy (public member function)
	`disable_recursion_pending`	disables recursion until the next increment (public member function)

`pop([ec])`: discard following iterations in subtree and go to next entry in the last level. This operation will invalidate all previous copies; and when `.depth() == 0`, iterator will become `end()`.

`disable_recursion_pending`: when the current entry is directory, do not go into the directory for next `++`. `recursion_pending` will return `false` then. Increment will reset `recursion_pending` to `true`.

Directory iteration



Directory iteration

- Note 1: their `operator*` returns `const&`, so you cannot modify `directory_entry` without copying it.
- Note 2: they are input iterators.
  - Recap: input iterators are not forward iterators because they're one-pass; copying it and iterating the copy may lead to different results.
  - Here similarly, directory contents may be altered by other processes, so it's not multi-pass.
    - It's actually more complicated and **see our homework** for detailed discussion.
- Note 3: if new files and directories are added in directory `A` after creation of `(recursive_)directory_iterator{ "A" }`, it's unspecified whether they will be iterated or not.

Final Notes

- There exist some other minor read-only methods:
  1. Path operations:
     - `current_path() -> p` for CWD;
     - `temp_directory_path() -> p` (e.g. `/tmp` on Linux);
     - `equivalent(p1, p2) -> bool` for judging whether two paths are essentially same (following symlinks).
     - `read_symlink(p1) -> p2` for converting a symlink path to its target path (error if `p1` is not symlink).
  2. Filesystem operations:
     - `is_empty(p) -> bool`, return `true` if `p` is empty file or directory;
     - `space(p) -> space_info`, which determines space of mounted filesystem that `p` lies in.
       - On Linux, you can check filesystem of the path by `df -h`:

```
jiaming@jiaming-Legion:~/Self/test_2026$ df -h / /home /tmp /dev
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p7  182G  133G   40G  78% /
/dev/nvme0n1p7  182G  133G   40G  78% /
/dev/nvme0n1p7  182G  133G   40G  78% /
udev            7.8G     0  7.8G   0% /dev
```

       - On Windows, each drive is seen as a mounted filesystem.

Final Notes

Capacity	Free	Available	Use%	Dir
8,298,082,304	8,298,082,304	8,298,082,304	0	/dev/null
194,491,936,768	52,302,696,448	42,348,515,328	77	/tmp
194,491,936,768	52,302,696,448	42,348,515,328	77	/home
0	0	0	100	/proc
194,491,936,768	52,302,696,448	42,348,515,328	77	/

- An example program from cppreference:

```
struct space_info {
    std::uintmax_t capacity;
    std::uintmax_t free;
    std::uintmax_t available;
};
```

capacity	total size of the filesystem, in bytes (public member object)
free	free space on the filesystem, in bytes (public member object)
available	free space available to a non-privileged process (may be equal or less than free) (public member object)

```
std::uintmax_t disk_usage_percent(const std::filesystem::space_info& si) noexcept
{
    if (constexpr std::uintmax_t X(-1);
        si.capacity == 0 || si.free == 0 || si.available == 0 ||
        si.capacity == X || si.free == X || si.available == X
    )
        return 100;

    std::uintmax_t unused_space = si.free, capacity = si.capacity;
    const std::uintmax_t used_space{capacity - unused_space};
    return 100 * used_space / capacity;
}

void print_disk_space_info(auto const& dirs, int width = 14)
{
    std::cout << std::left;
    for (const auto s : {"Capacity", "Free", "Available", "Use%", "Dir"})
        std::cout << "| " << std::setw(width) << s << ' ';

    for (std::cout << '\n'; auto const& dir : dirs)
    {
        std::error_code ec;
        const std::filesystem::space_info si = std::filesystem::space(dir, ec);
        for (auto x : {si.capacity, si.free, si.available, disk_usage_percent(si)})
            std::cout << "| " << std::setw(width) << static_cast<std::intmax_t>(x) << ' ';
        std::cout << "| " << dir << '\n';
        std::cout << ec.message() << "\n";
    }
}
```

Supplementary

- File system
  - Path operations
  - File system operations
    - Overview
    - File status query and directory iteration
    - Modification operations

Creation

- There are four kinds of modification operations:
  1. Create:

`create_directory` (C++17) `create_directories` (C++17)	creates new directory (function)
`create_hard_link` (C++17)	creates a hard link (function)
`create_symlink` (C++17) `create_directory_symlink` (C++17)	creates a symbolic link (function)

① directory:

- `create_directory(p[, existing_p[, ec]]) -> bool;`
  - When given `existing_p`, created `p` will copy OS-dependent attributes from another directory `existing_p` (Windows does nothing). Parent of `p` must exist (i.e. create only a single directory).
- `create_directories(p[, ec]) -> bool;`
  - Create every element that doesn't exist in directory path. So when parent of `p` doesn't exist, it will be created.

return `false` if directory already exists (not seen as error) or an error occurs.

Creation

ATTENTION: what the path symlink refers to is from the view of symlink itself. (but hard link doesn't do so).

② links:

- `create_hard_link(original_path, link_path[, ec]);`
- `create_symlink(original_path, link_path[, ec]);`
- `create_directory_symlink(original_path, link_path[, ec]);`
  - Some OS may need distinct APIs to create symbolic links to regular files and directories, so it's recommended to use `create_directory_symlink` for directory. POSIX doesn't distinguish them.
  - Note: these two methods just create soft links, which then require root privilege in Windows. And, there is no way to create junction in Windows by standard library.

③ Regular files: not in `stdfs`; you can use `std::ofstream` to do so.

Example:

```
fs::create_directories("sandbox/subdir");
fs::create_symlink("target", "sandbox/sym1");
fs::create_directory_symlink("subdir", "sandbox/sym2");

for (auto it = fs::directory_iterator("sandbox"); it != fs::directory_iterator(); ++it)
    if (is_symlink(it->symlink_status()))
        std::cout << *it << "->" << read_symlink(*it) << '\n';

assert(std::filesystem::equivalent("sandbox/sym2", "sandbox/subdir"));
```

```
"sandbox/sym1"->"target"
"sandbox/sym2"->"subdir"
```

Copy

2. Copy:

`copy` (C++17)	copies files or directories (function)
`copy_file` (C++17)	copies file contents (function)
`copy_symlink` (C++17)	copies a symbolic link (function)

- Copy supports `copy_options` (again, bit-operatable scoped enumeration) to configure copy behavior.
- `copy(from, to[, opt[, ec]])`: copy according to `opt`.
- `copy_file(from, to[, opt[, ec]])`: copy regular file according to `opt`.
  - This can be cheaper than `copy` for less check, and doesn't incur TOC/TOU compared with `if(!is_directory(a)) copy(a, b)`.
  - Error conditions (after following symbolic links, don't cover OS error like permission):
    - `!is_regular_file(from)`; or
    - `exists(to) && (equivalent(from, to) || !is_regular_file(to) || opt == copy_options::none)`.

The specific behavior (including error conditions) for `copy` is too long and thus not covered here. Check details in cppreference.

Copy option enumerators

At most one copy option in each of the following options groups may be present, otherwise the behavior of the copy functions is undefined.

Member constant	Meaning
options controlling `copy_file()` when the file already exists	
`none`	Report an error (default behavior).
`skip_existing`	Keep the existing file, without reporting an error.
`overwrite_existing`	Replace the existing file.
`update_existing`	Replace the existing file only if it is older than the file being copied.
options controlling the effects of `copy()` on subdirectories	
`none`	Skip subdirectories (default behavior).
`recursive`	Recursively copy subdirectories and their content.
options controlling the effects of `copy()` on symbolic links	
`none`	Follow symlinks (default behavior).
`copy_symlinks`	Copy symlinks as symlinks, not as the files they point to.
`skip_symlinks`	Ignore symlinks.
options controlling the kind of copying `copy()` does	
`none`	Copy file content (default behavior).
`directories_only`	Copy the directory structure, but do not copy any non-directory files.
`create_symlinks`	Instead of creating copies of files, create symlinks pointing to the originals. Note: the source path must be an absolute path unless the destination path is in the current directory.
`create_hard_links`	Instead of creating copies of files, create hardlinks that resolve to the same files as the originals.

(Also applied to copy for files).

Copy

- `copy_symlink(from, to[, ec])`: copy symbolic link to another position, instead of following link and copy content like `copy`.
  - Equiv. to `create_(directory_)symlink(read_symlink(from), to)`.
  - Note: `copy_symlink` doesn't copy junction on Windows; you can only use `copy(from, to, copy_options::copy_symlinks)` to copy it.
- Example of copy:

```
fs::create_directories("sandbox/dir/subdir");
std::ofstream("sandbox/file1.txt").put('a');
fs::copy("sandbox/file1.txt", "sandbox/file2.txt"); // copy file
fs::copy("sandbox/dir", "sandbox/dir2"); // copy directory (non-recursive)
const auto copyOptions = fs::copy_options::update_existing
                       | fs::copy_options::recursive
                       | fs::copy_options::directories_only
                       ;
fs::copy("sandbox", "sandbox_copy", copyOptions);
static_cast<void>(std::system("tree"));
fs::remove_all("sandbox");
fs::remove_all("sandbox_copy");
```

```
.
├── sandbox
│   ├── dir
│   │   └── subdir
│   ├── dir2
│   ├── file1.txt
│   └── file2.txt
└── sandbox_copy
    ├── dir
    │   └── subdir
    └── dir2

8 directories, 2 files
```

Remove and Rename

3. Remove:
   - `remove(p[, ec]) -> bool`: requires `p` to be regular file or empty directory, symlink is not followed; return `true` if `p` is removed.
   - `remove_all(p[, ec]) -> uintmax_t`: remove all contents in `p` (including `p`), symlink is not followed; return number of removed entries (-1 on error).
4. Rename:
   - `rename(old_p, new_p[, ec])`:
     - If `new_p` is hard link to `old_p`, nothing happens;
     - If `old_p` is regular file: remove `new_p` if it exists and is a regular file, and rename to `new_p`.
     - If `old_p` is directory: remove `new_p` if it exists and is a directory, and rename to `new_p`. Parent of `new_p` should exist and `new_p` shouldn't end with separator.
     - Note: other processes cannot observe such removal.